

CompTIA.

# A+ Core 2

Exam 220-1202

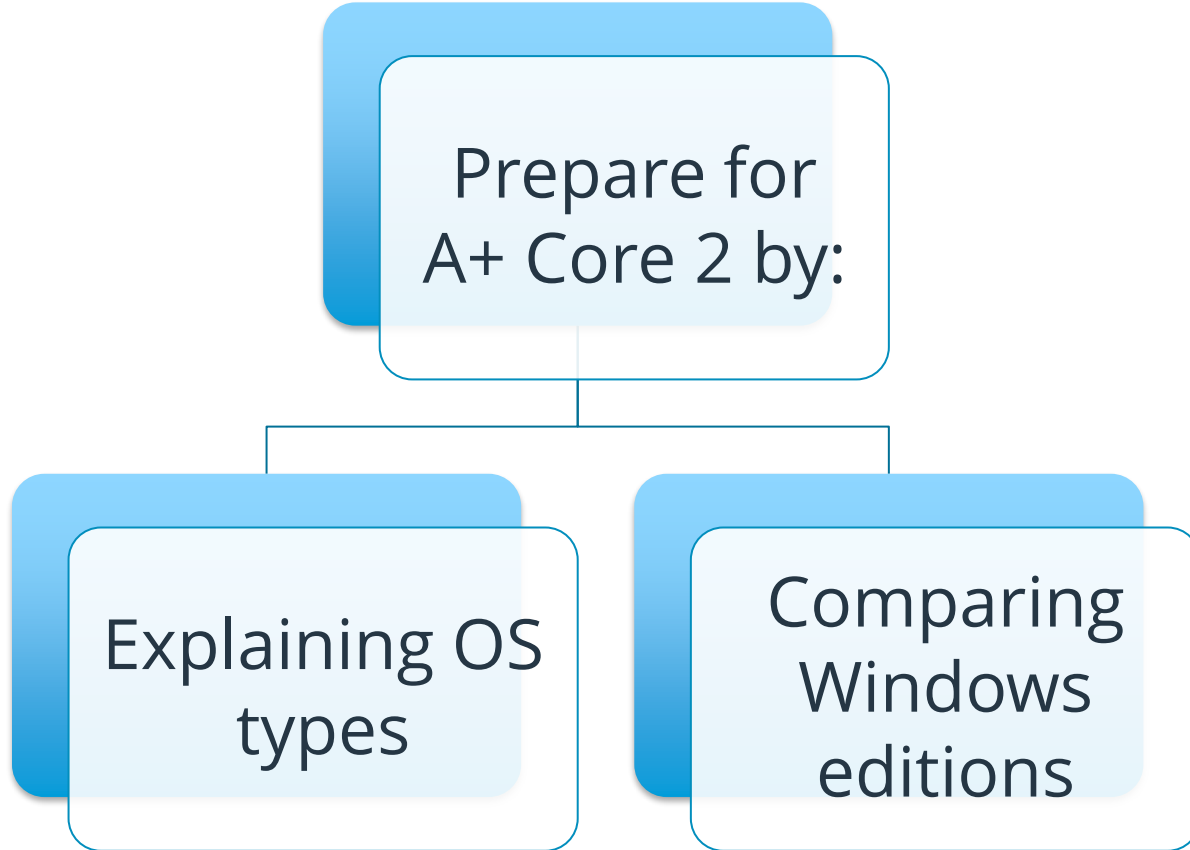
A+ 220-1202

# Module 16



## Installing Operating Systems

# Learning Outcomes



# Lesson 16.1

## Windows Editions

# Windows Versions

32-bit vs 64-bit

- Windows 11 only in 64-bit edition

Home, Pro, Pro for  
Workstations,  
Enterprise, Education

N versions compliant  
with EU regulations

# Windows Home Edition



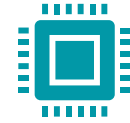
**Designed for SOHO  
use**



**Internet connection  
required for Microsoft  
account login and  
activation**



**Licensing  
OEM  
Retail**



**Limitations  
64 cores  
Single CPU  
128 GB RAM (64-bit)**

# Work and Education Features

Pro, Enterprise, and Education editions

Domain support

Group Policy Editor

BitLocker

- Disk encryption for hard drives and USB (not in Home edition)

RDP

- Client and server (Home only has RDP Client)

# Windows Pro and Enterprise Editions (Slide 1 of 2)

Pro

- OEM, retail, volume licensing
- Pro for Workstations is same as Pro but support for larger memory and advanced hardware

Enterprise and Education

- Volume licensing only
- DirectAccess VPN, AppLocker, MS Desktop Optimization Pack

128 core max (Pro and Education)

256 core max (Pro for Workstations and Enterprise)

2TB RAM (Pro and Education)

6TB RAM (Pro for Workstations and Enterprise)



# Windows Pro and Enterprise Editions (Slide 2 of 2)

System > About

Device specifications

Rename this PC

Copy

|                  |                                                         |
|------------------|---------------------------------------------------------|
| Device name      | WORKSUB01                                               |
| Full device name | WORKSUB01                                               |
| Processor        | 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 1.38 GHz |
| Installed RAM    | 8.00 GB (7.75 GB usable)                                |
| Device ID        | 8442D1A-8                                               |
| Product ID       | 00000-00000-00000-00000-00000                           |
| System type      | 64-bit operating system, x64-based processor            |
| Pen and touch    | No pen or touch input is available for this display     |

Related links: [Domain or workgroup](#) [System protection](#) [Advanced system settings](#)

Windows specifications

Copy

|              |                                                   |
|--------------|---------------------------------------------------|
| Edition      | Windows 11 Enterprise                             |
| Version      | 22H2                                              |
| Installed on | 6/30/2023                                         |
| OS build     | 22621.4169                                        |
| Experience   | Windows Feature Experience Pack 1000.22700.1034.0 |

[Microsoft Services Agreement](#)  
[Microsoft Software License Terms](#)

Screenshot courtesy of Microsoft.

# Windows Upgrade Paths and Feature Updates



## From Windows 10 to 11

TPM 2.0, UEFI with Secure Boot,  
supported processor

Upgrade with same edition or  
from Home to Pro

Need license to update from  
Home to Enterprise

Downgrade from Pro to Home



## Win 10

Semi-annual feature updates  
Microsoft support ends Oct 2025



## Win 11

Annual feature update cycle

## Lab Activity

- Lab: Windows 11 Features/Desktop  
(available in CertMaster Learn and CertMaster Perform)

## Lesson Review: Windows Editions

- Windows Versions
- Windows Home Edition
- Work and Education Features
- Windows Pro and Enterprise Editions
- Windows Upgrade Paths and Feature Updates

# Lesson 16.2

## OS Installations and Upgrades

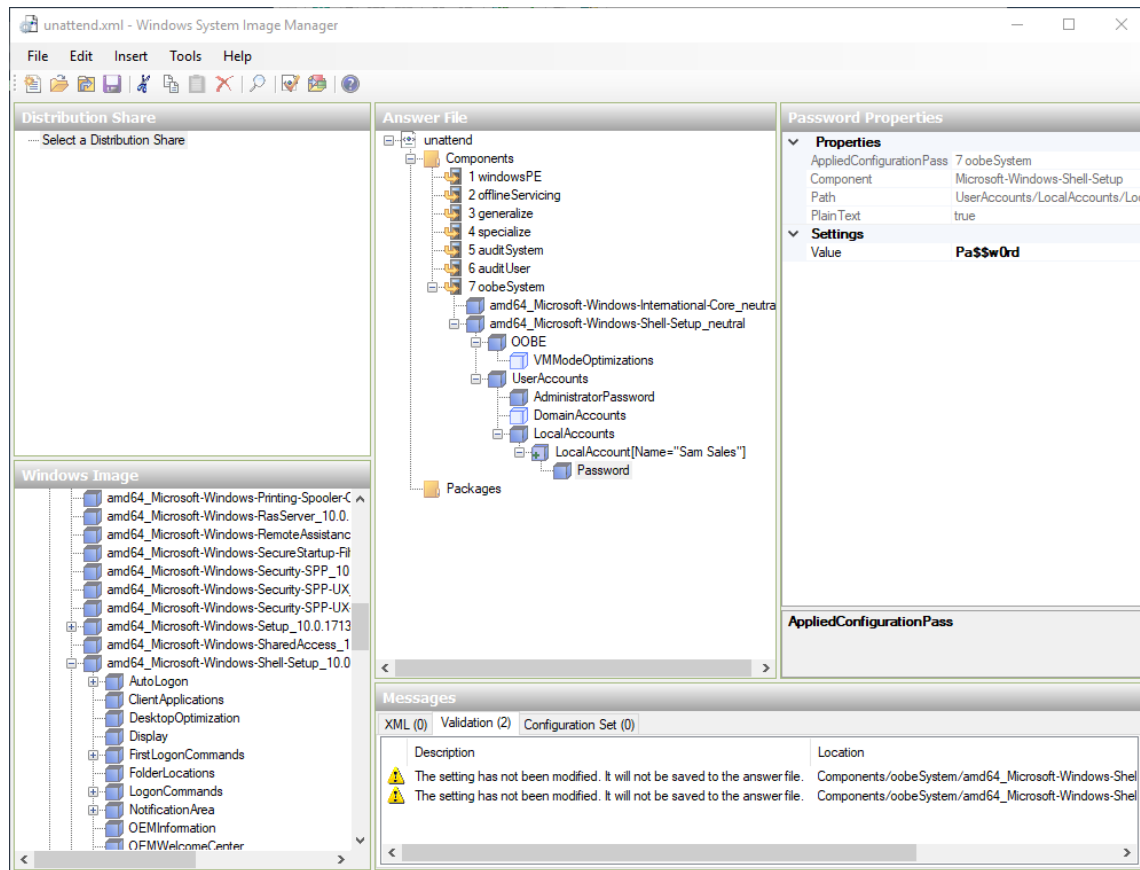
# Installation and Upgrade Considerations

- Clean install
  - Repartition and reformat drive then install
- In-place upgrade
  - Updates existing OS install with new OS
  - User data remains on system
- Considerations
  - Hardware compatibility
  - Application and driver support
  - Backup files and user preferences
  - Third-party drivers
- Feature updates

# Unattended Installations (Slide 1 of 2)

- Image deployment with answer file
- Network-pushed installation
  - Zero-touch install
    - Remote install using cloud-based service to deploy the image onto hardware

# Unattended Installations (Slide 2 of 2)



Screenshot courtesy of Microsoft.



# Boot Devices (Slide 1 of 2)

- Configured in BIOS/UEFI config
- Optical media
- USB, external drive, flash drive, hot-swappable drive
- Network boot
  - PXE boot (pronounced "pixie")
  - DHCP server required
- Internet-based boot
- Internal drive
- Multiboot
  - Multiple OSs installed on their own partition
  - OS selection menu shows up during boot

# Boot Devices (Slide 2 of 2)

| Main                                                             | Security | Advanced  | UEFI Drivers |
|------------------------------------------------------------------|----------|-----------|--------------|
| <b>← Boot Options</b>                                            |          |           |              |
| Startup Menu Delay (sec.)                                        |          | 5         |              |
| <input checked="" type="checkbox"/> Fast Boot                    |          |           |              |
| <input checked="" type="checkbox"/> CD-ROM Boot                  |          |           |              |
| <input checked="" type="checkbox"/> USB Storage Boot             |          |           |              |
| <input checked="" type="checkbox"/> Network (PXE) Boot           |          |           |              |
| After Power Loss                                                 |          | Power off | ?            |
| <input checked="" type="checkbox"/> Prompt on Memory Size Change |          |           |              |
| <input type="checkbox"/> Prompt on Fixed Storage Change          |          |           |              |
| <input checked="" type="checkbox"/> Audio Alerts During Boot     |          |           |              |
| <input checked="" type="checkbox"/> UEFI Boot Order              |          |           |              |
| -----                                                            |          |           |              |
| M.2: Windows Boot Manager                                        |          |           |              |
| Save                                                             |          | Help      | Exit         |

# Disk Configuration (Slide 1 of 2)

## MBR-style partition

- Master boot record takes first 512-byte to store the partition table

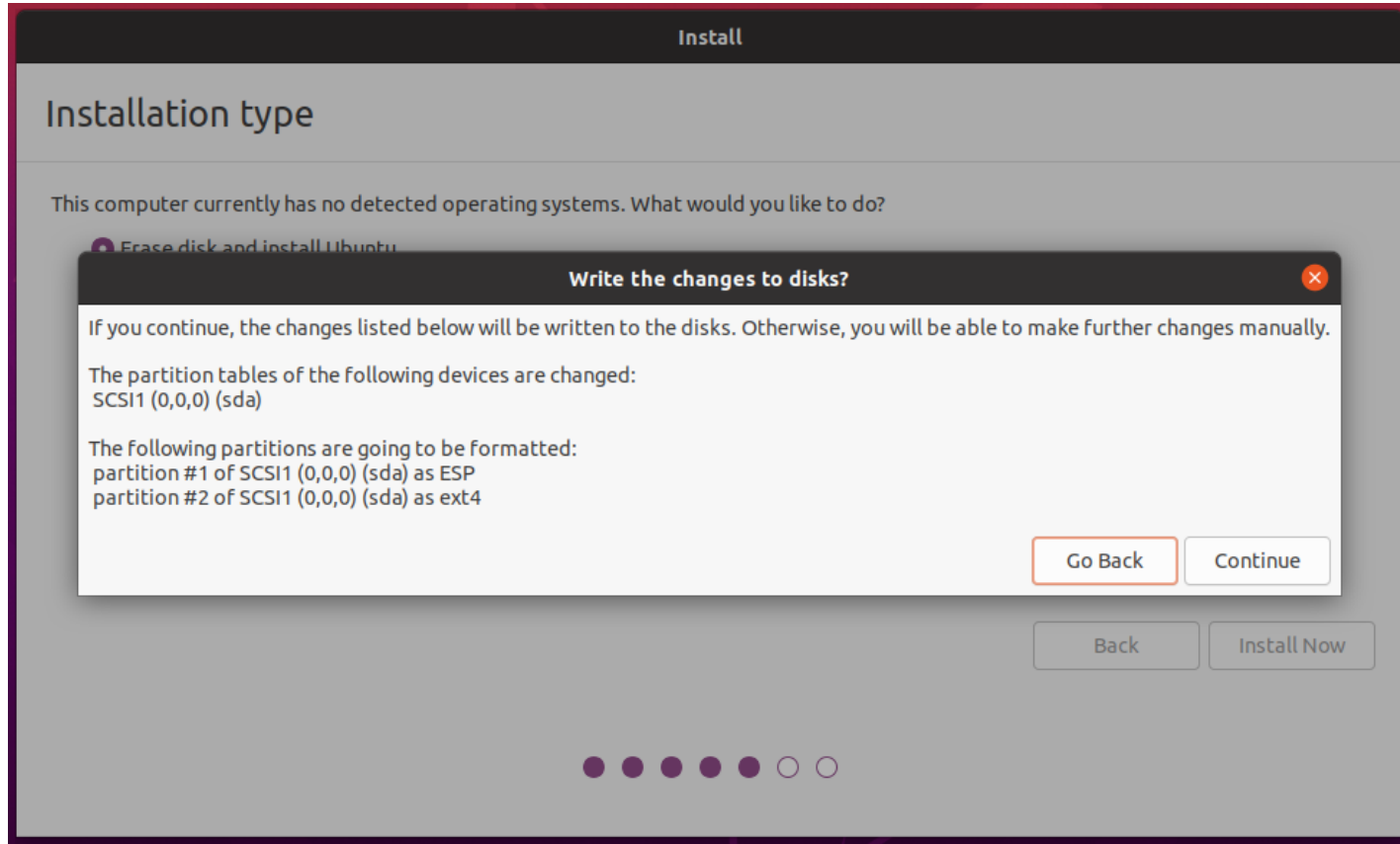
## GPT-Style partition

- GUID partition table
- Larger number of partitions (128 in Windows) and larger partition size (over 2TB)

## Drive Format

- NTFS for Windows
- APFS for macOS
- ext3/4 for Linux

# Disk Configuration (Slide 2 of 2)



# Repair Installation

## Recovery partition

- OEM created on drive

## Reset Windows

- Refresh reinstalls system files and resets most settings to default
- Reset erases all user data and resets the system for fresh installation

## Lab Activity

- Applied Live Lab: Perform a Remote Network Installation (available in CertMaster Perform and CertMaster Labs)
- Challenge Live Lab: Support Windows Installation and Upgrade Issues (available in CertMaster Perform and CertMaster Labs)

## Lesson Review: OS Installations and Upgrades

- Installation and Upgrade Considerations
- Unattended Installations
- Boot Devices
- Disk Configuration
- Repair Installation